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Networking module 5b

Link layer: Ethernet & Switches
Wireless networking

CREATING TOMORROW

Link layer

Ethernet

Switches

- switching tables

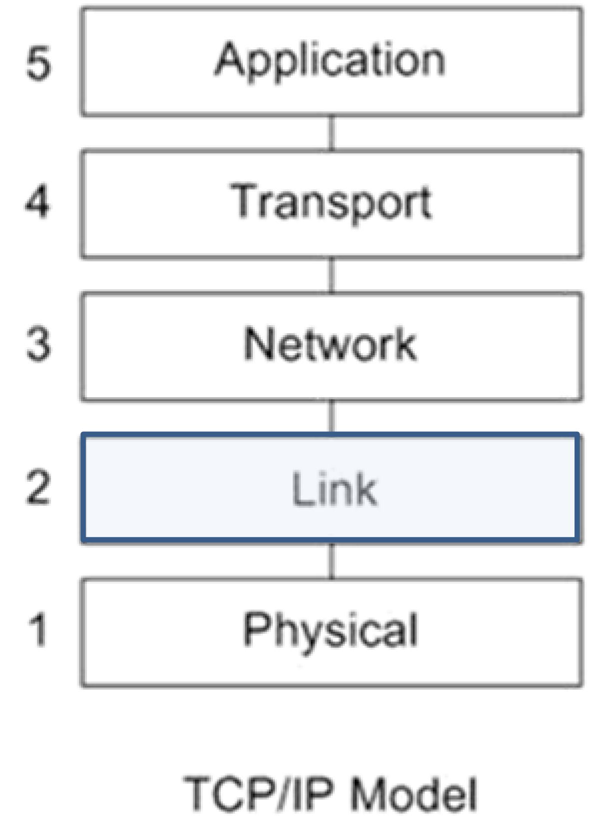

Figure 1

Link layer: reminder

- Link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link

- Uses MAC addresses, e.g.: 1A-2F-BB-76-09-AD

- Hosts use ARP queries to find MAC address for a certain IP address

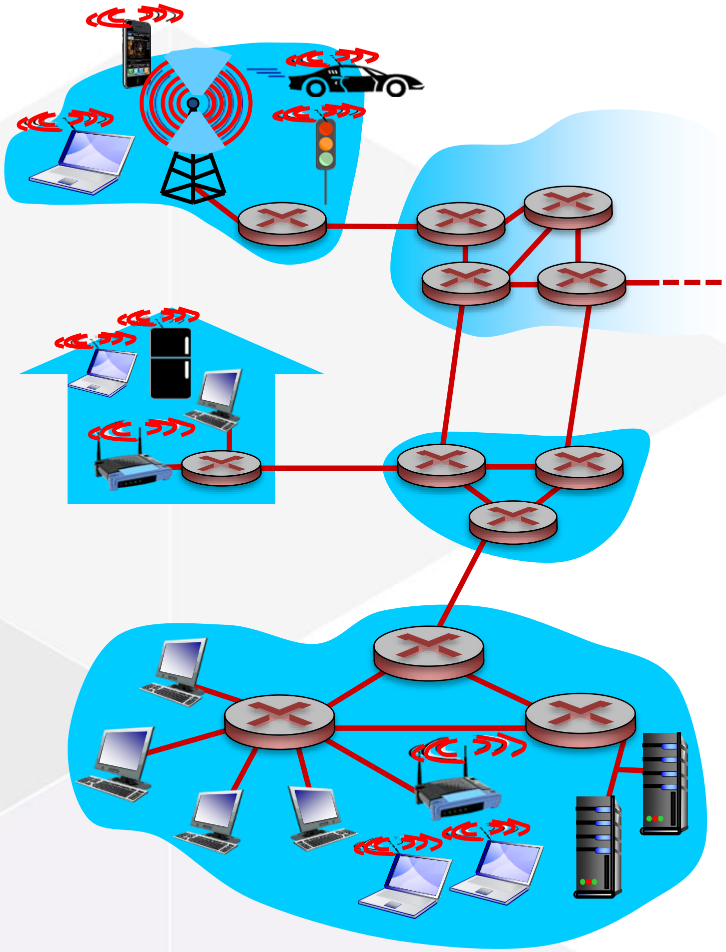

Figure 2

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Ethernet

Figure 24

Ethernet

- "dominant" wired LAN technology
- simple, cheap
- kept up with speed race: 10 Mbps – 10 Gbps

Ethernet frame

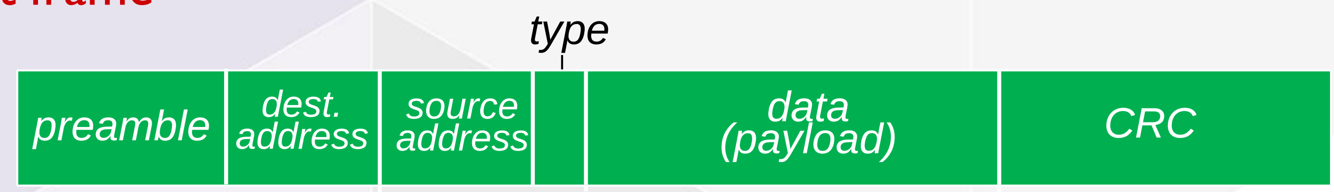

Figure 23

- Preamble: fixed pattern used to synchronise sender, receiver clocks
- Source and Destination MAC address
- Type: usually IP
- Data: contains IP datagram
- CRC: cyclic redundancy check code to detect errors (corrupt frames are dropped)

Ethernet and switches

Collision domains

- One host can send at the same time, hosts need to wait when line is busy
- When two hosts start sending at the same time, data collides and becomes corrupted.
- All of this decreases performance

Bus topology: popular through mid 90s

- all nodes in same collision domain

Star topology: prevails today

- active *switch* in center
- each "spoke" runs a (separate) Ethernet (nodes do not collide with each other)

Figure 10

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Switch: *multiple* simultaneous transmissions

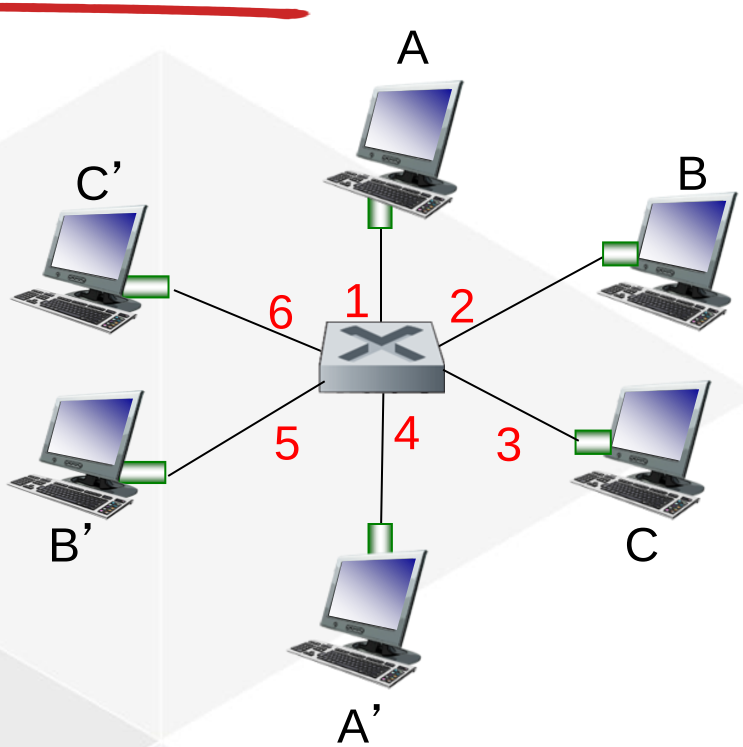
hosts have dedicated, direct connection to switch

switches buffer packets

Ethernet protocol used on *each* incoming link, but no collisions; full duplex

- each link is its own collision domain

switching: A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces (1,2,3,4,5,6)

Figure 11

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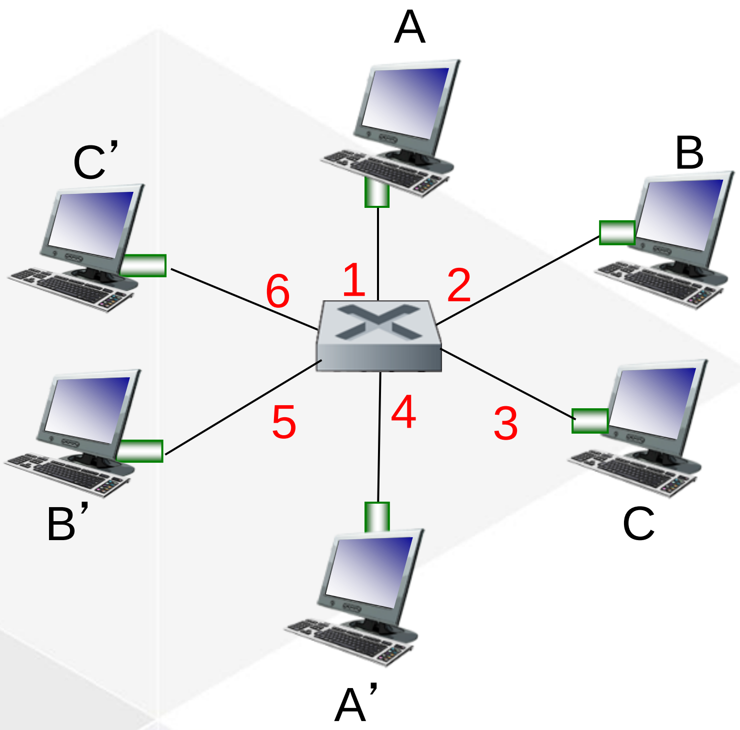
Switch forwarding table

Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

- A: *each switch has a switch table, each entry:*
  - *(MAC address of host, interface to reach host, time stamp)*
  - *looks like a routing table!*

Q: how are entries created, maintained in switch table?



switch with six interfaces (1,2,3,4,5,6)

Figure 11

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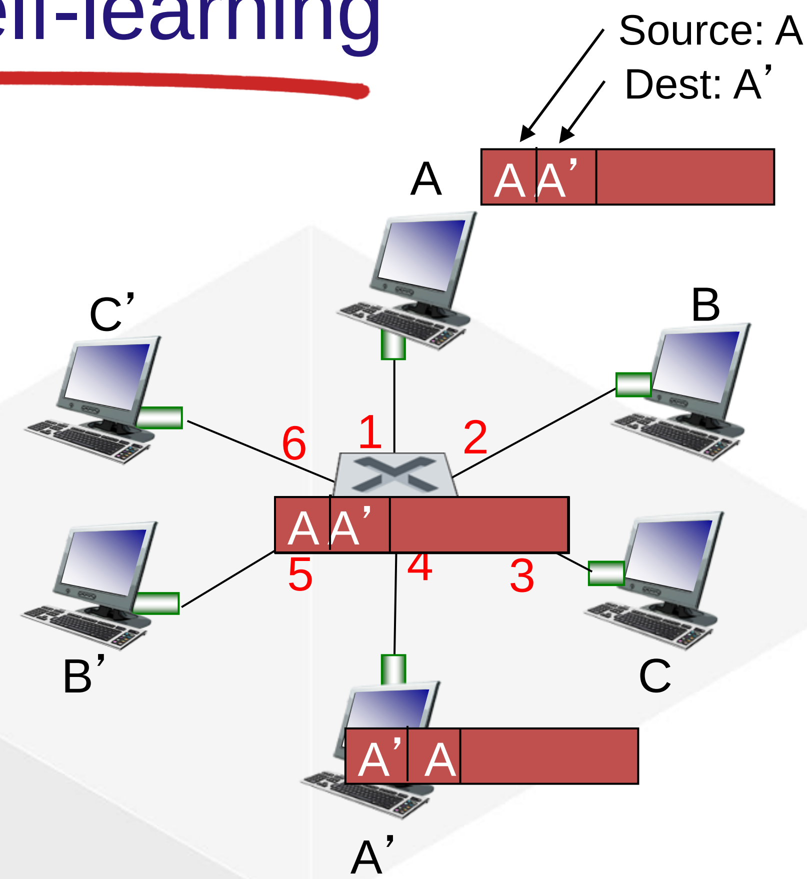
Switch: self-learning

switch *learns* which hosts can be reached through which interfaces

- when frame received, switch "learns" location of sender
- records sender/location pair in switch table
- switches do **NOT** use ARP

frame destination, A', location unknown: *flood*

destination A location known: *selectively send on just one link*



MAC addr	interface	TTL
A	*1*	*60*
A'	*4*	*60*

switch table - initially empty

Switches vs. Routers

both are store-and-forward:

- *routers:* network-layer devices
  - examine network-layer headers
  - connect different subnets
- *switches*: link-layer devices
  - examine link-layer headers
  - connect devices on same subnet

both have forwarding tables:

- *routers:* compute tables using routing algorithms, IP addresses
- *switches:* learn forwarding table using MAC addresses

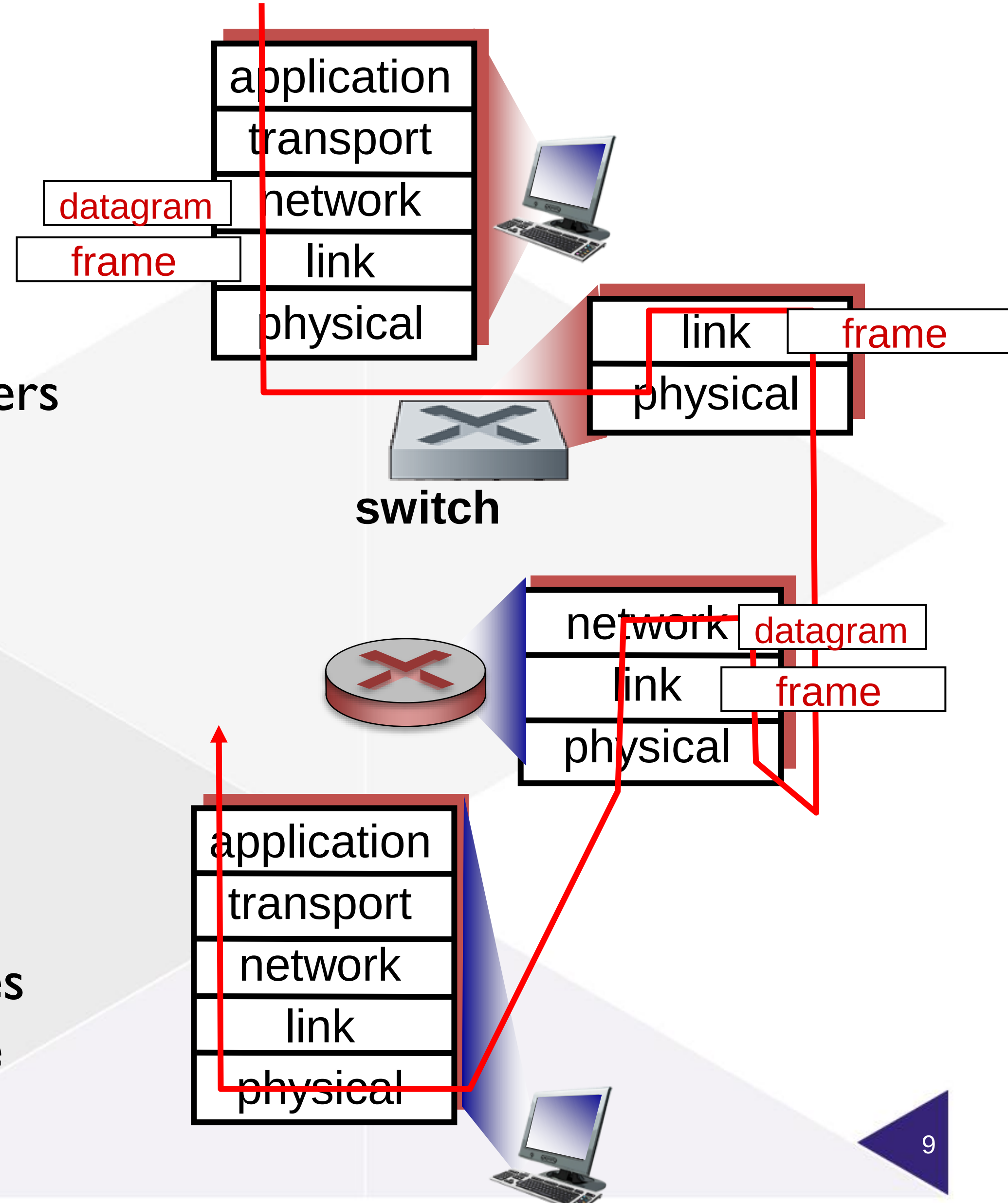


Figure 13

Wireless networks

Elements of a wireless network

- Wireless hosts
- Base stations
- Wireless links

Wireless versus mobile

Properties of wireless links

802.11 (WiFi)

Security in WiFi Networks

- WiFi Encryption
- Hacking WiFi
- WiFi encryption vs HTTPS

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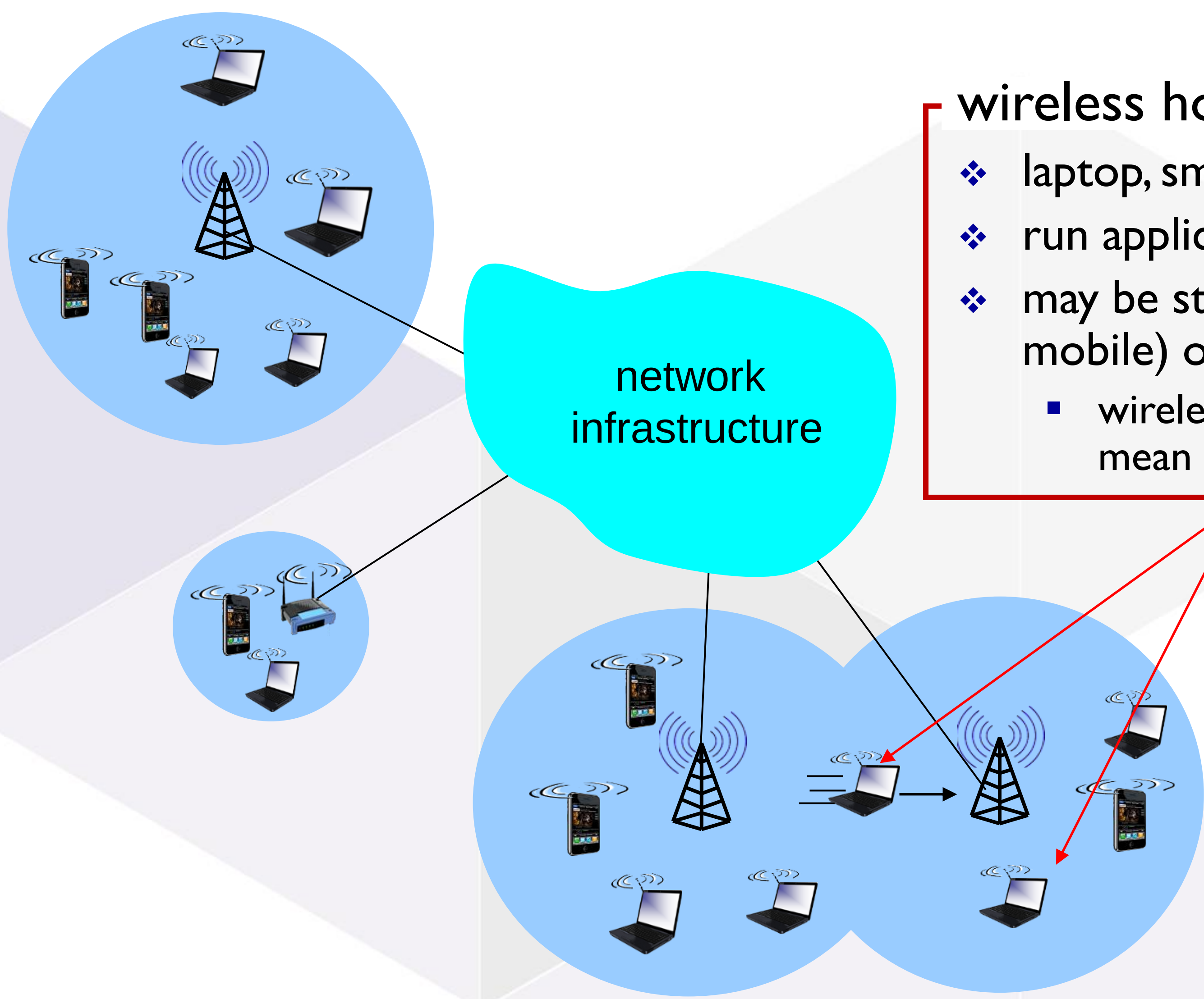
Elements of a wireless network



wireless hosts

- laptop, smartphone
- run applications
- may be stationary (non-mobile) or mobile
  - wireless does *not* always mean mobility

Figure 17

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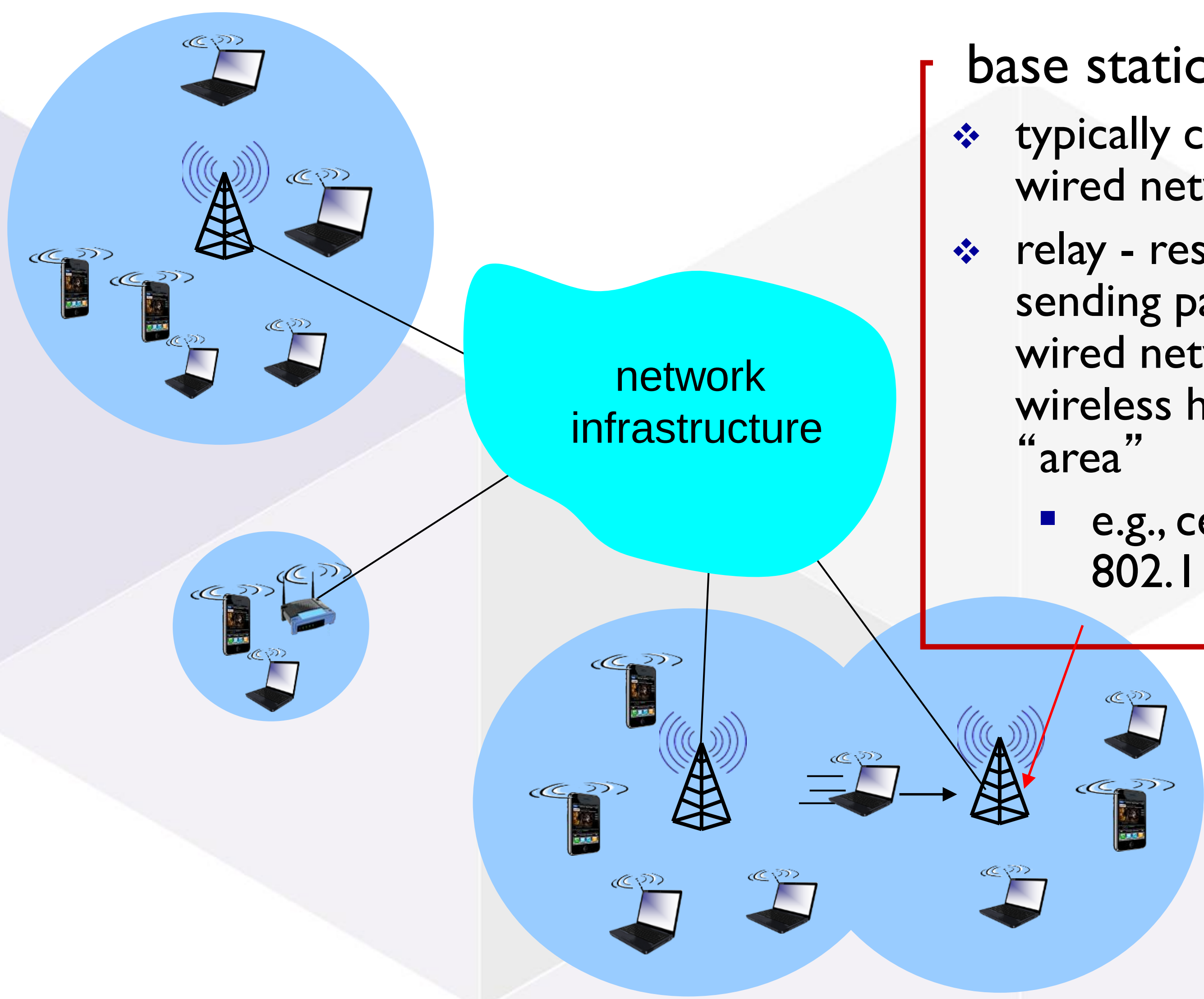
Elements of a wireless network



base station

- typically connected to wired network
- relay - responsible for sending packets between wired network and wireless host(s) in its "area"
  - e.g., cell towers, 802.11 access points

Figure 17

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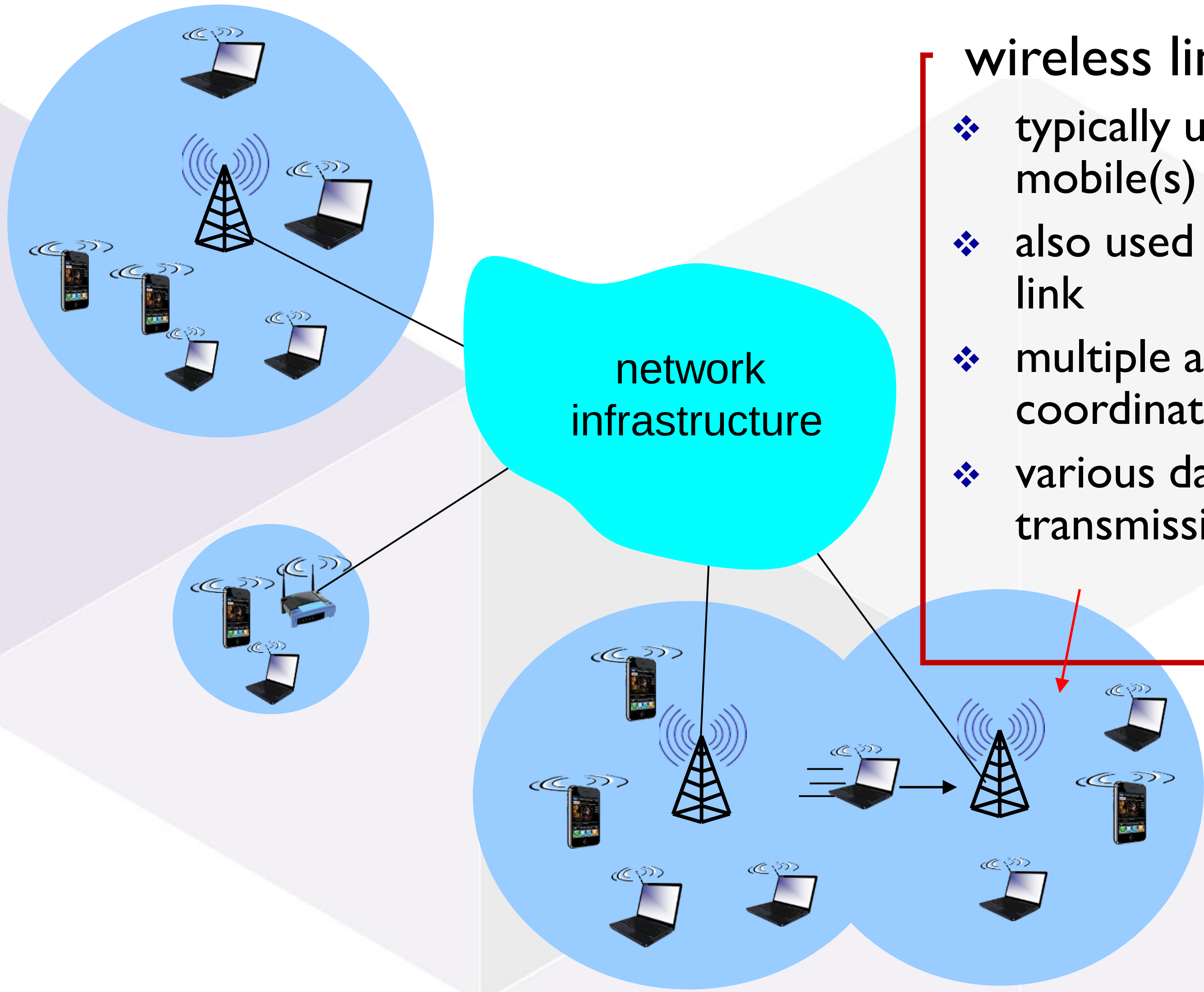
Elements of a wireless network



wireless link
- typically used to connect mobile(s) to base station
- also used as backbone link
- multiple access protocol coordinates link access
- various data rates, transmission distance

Figure 17

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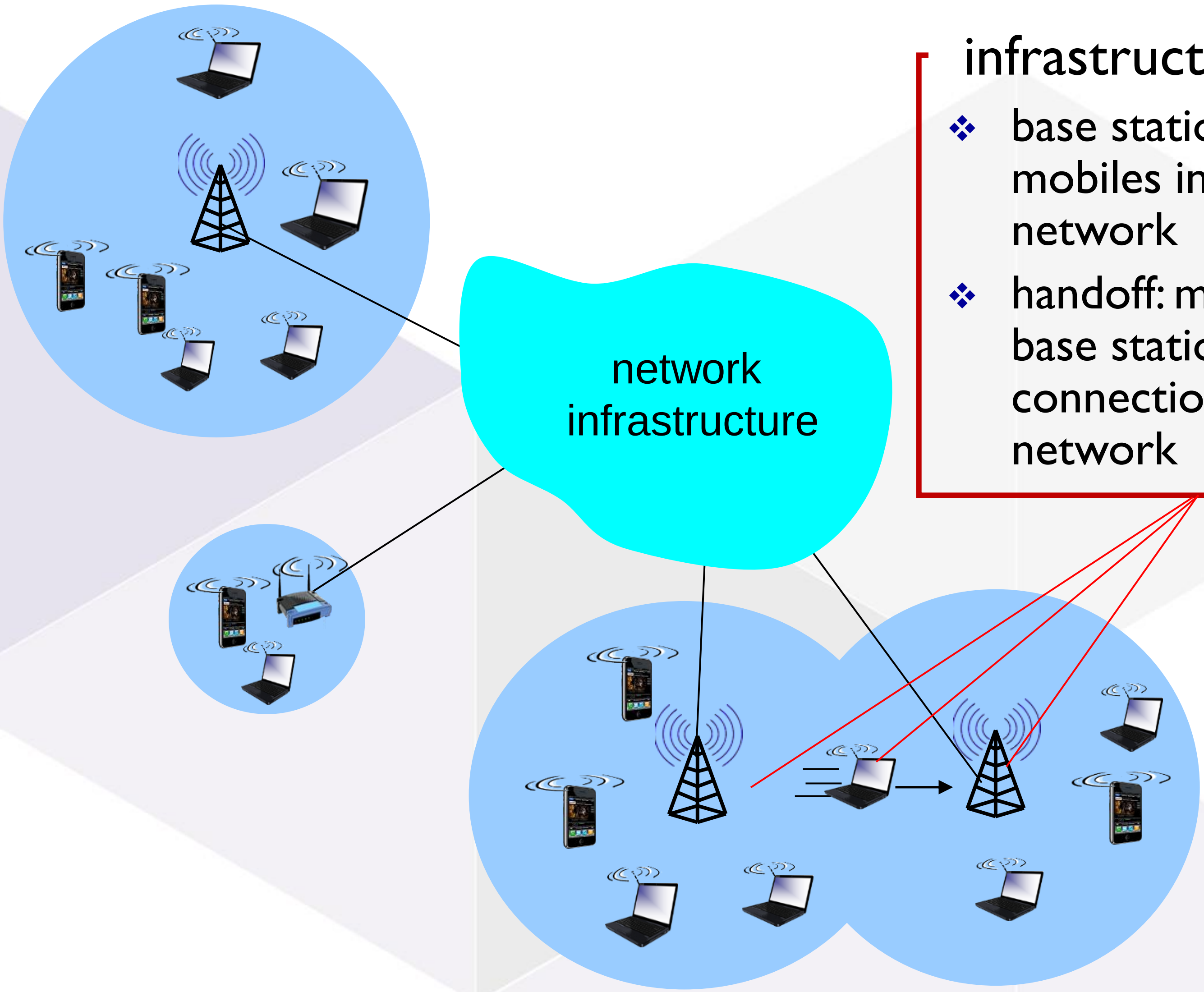
Elements of a wireless network



infrastructure mode

- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

Figure 17

What is mobility?

spectrum of mobility, from the *network* perspective:

no mobility

high mobility

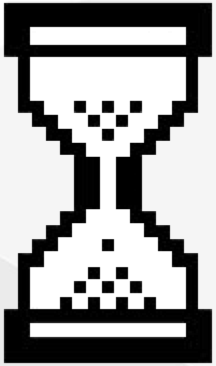
mobile wireless user, using same access point

mobile user, connecting/ disconnecting from network using DHCP.

mobile user, passing through multiple access point while maintaining ongoing connections (like cell phone)

Figure 24

Break



Characteristics of selected wireless links

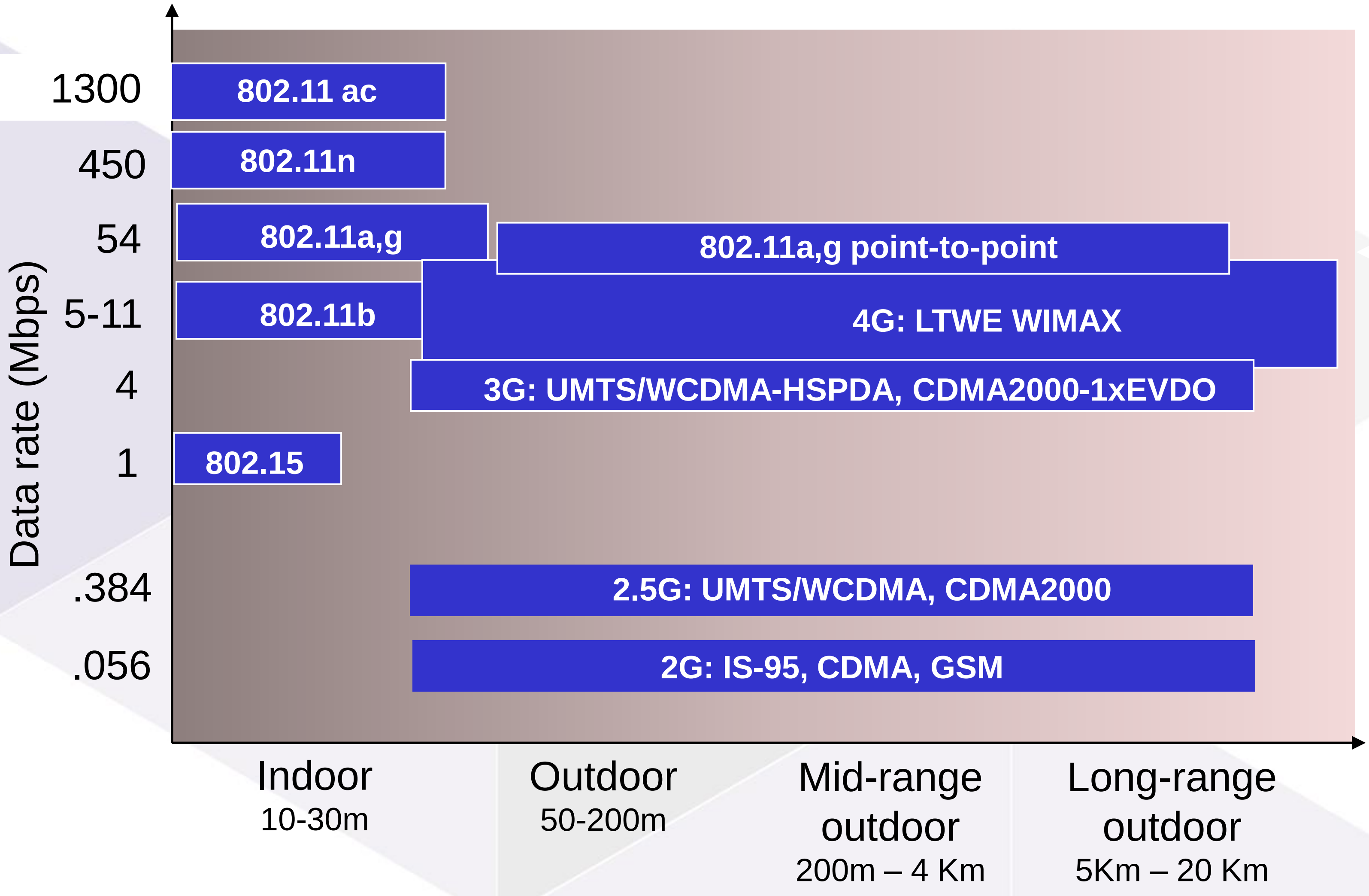


Figure 18

Wireless Link Characteristics

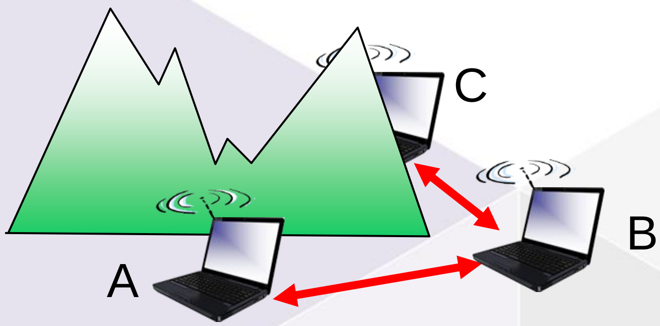
important differences from wired link ….

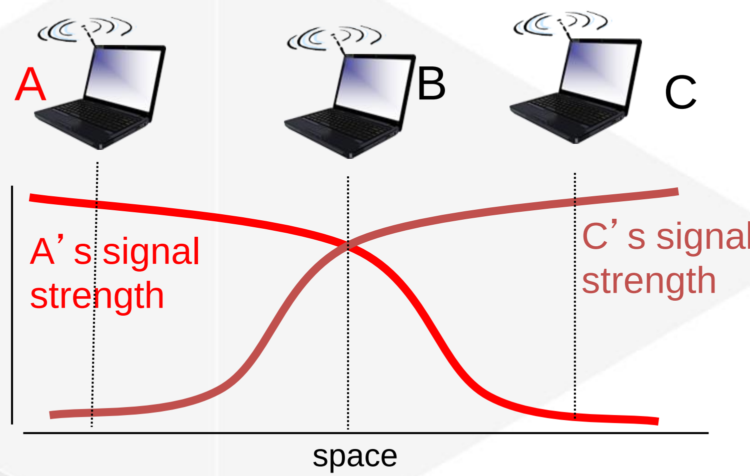
- *decreased signal strength:* radio signal attenuates as it propagates through matter (path loss)
- *interference from other sources:* standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- *multipath propagation:* radio signal reflects off objects ground, arriving ad destination at slightly different times

…. make communication across (even a point to point) wireless link much more "difficult"

Wireless network characteristics

Multiple wireless senders and receivers create additional problems (beyond multiple access):





Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B

Signal attenuation:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

Figure 19

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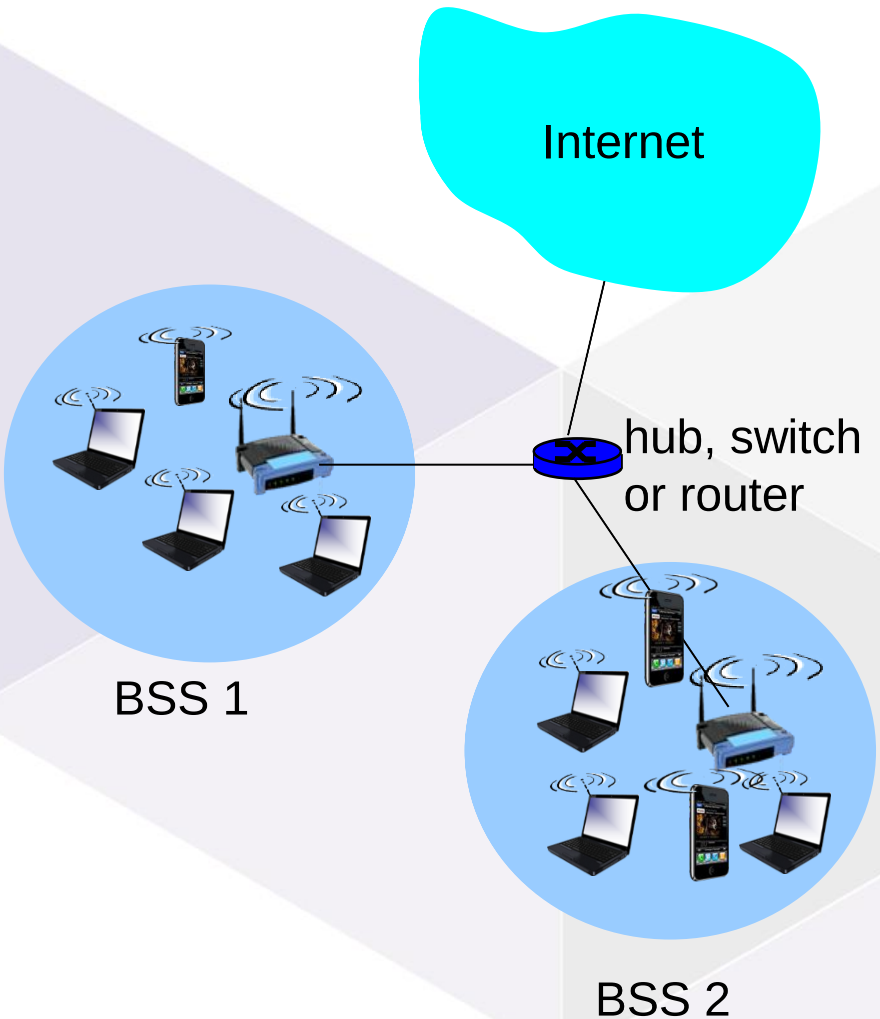
802.11 LAN architecture



- wireless host communicates with base station
  - base station = access point (AP)
- Basic Service Set (BSS) (aka "cell") in infrastructure mode contains:
  - wireless hosts
  - access point (AP): base station
  - ad hoc mode: hosts only

Figure 22

802.11: Channels, association

802.11b: 2.4GHz-2.485GHz spectrum divided into 11 channels at different frequencies

- AP admin chooses frequency for AP
- interference possible: channel can be same as that chosen by neighboring AP!

host: must *associate* with an AP

- scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
- selects AP to associate with
- may perform authentication
- will typically run DHCP to get IP address in AP's subnet

Afluisteren van draadloze verbindingen

- WiFi verbindingen gaan door de lucht en kunnen dus zonder veel moeite door iedereen afgeluisterd worden
- Er bestaan speciale WiFi cards voor dit doel
- Gecombineerd met een tool als Wireshark is alle verstuurde data zichtbaar, evenals info uit de gebruikte protocollen (bv IP en MAC adressen)
- Dus: gebruik encryptie!

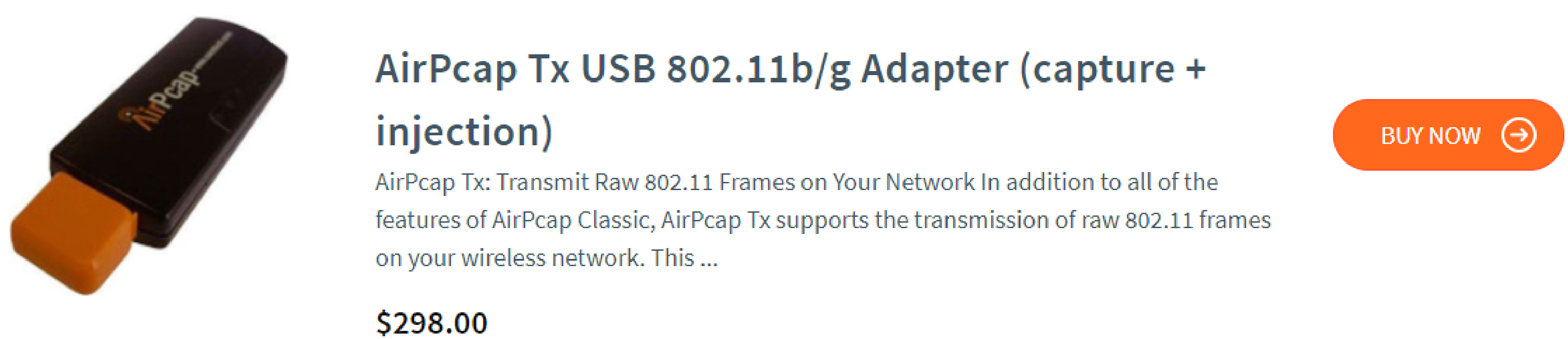


Figure 25

Review: Wifi authentication (& encryption)

WEP (1999) first 64-bit then 128-bit now 256-bit

- Security flaws since 2001, no longer secure since 2004

WPA (2003), 256-bit with

  - Integrity checks
  - TKIP: per packet key
  - AES superseded TKIP

- But roll-out on existing hardware meant: still vulnerable

WPA2 (2006) with

  - AES and CCMP Counter Cipher Mode with **Block Chaining** Message Authentication

- Known vulnerabilities: sustained attack through WPS (Wi-Fi Protected Setup) which must be disabled or flashed out of firmware
- Personal vs Enterprise: the latter uses an authentication server (such as Radius) to generate **certificates**.

Hacking WEP password in minutes

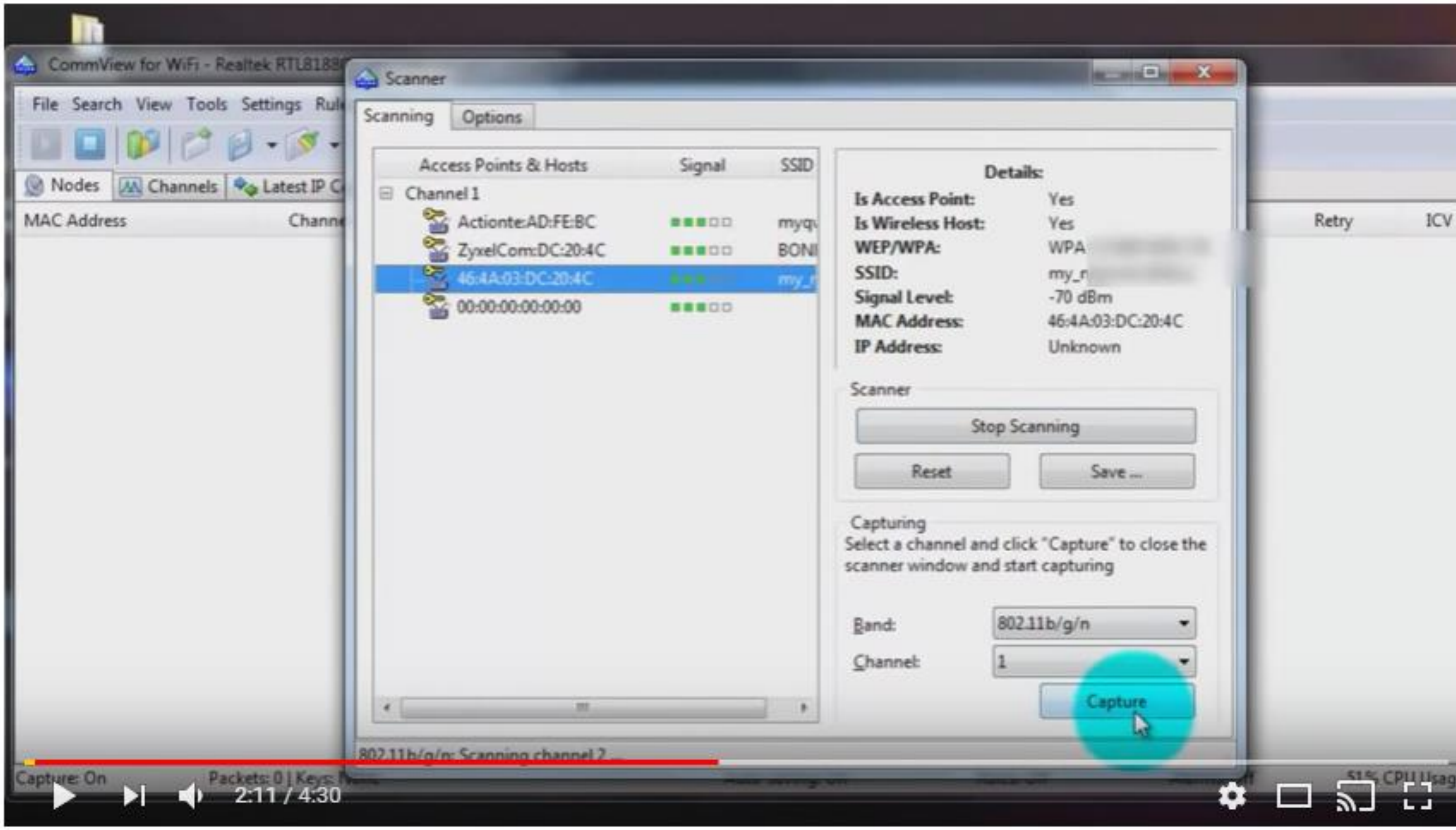


Figure 26

[Link to movie]

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WPA2 hacken: brute force

1. Luister het WiFi netwerk af, totdat je een bericht onderschept van iemand die zich aanmeldt
2. Dat bericht is versleuteld, dus je weet het wachtwoord nog niet
3. Probeer zelf (offline) aanmeldings-berichten te genereren waarbij je één voor één populaire wachtwoorden probeert.
4. Als je zelf gemaakte bericht overeenkomt met het afgeluisterde bericht, heb je het wachtwoord gevonden
5. Je kunt je nu aanmelden op het WiFi netwerk

Figure 28

RANK	PASSWORD	CHANGE FROM 2015
1	123456	Unchanged
2	password	Unchanged
3	12345	2 ↗
4	12345678	1 ↘
5	football	2 ↗
6	qwerty	2 ↘
7	1234567890	5 ↗
8	1234567	1 ↗
9	princess	12 ↗
10	1234	2 ↘
11	login	9 ↗
12	welcome	1 ↘
13	solo	10 ↗
14	abc123	1 ↘
15	admin	NEW
16	121212	NEW
17	flower	NEW
18	passw0rd	6 ↗
19	dragon	3 ↘
20	sunshine	NEW
21	master	4 ↘
22	hottie	NEW
23	loveme	NEW
24	zaq1zaq1	NEW
25	password1	NEW

The future: WPA3

- Available since 2018 (but not much used yet)
- Ad-hoc encryption for open WiFi hotspots (offers encryption for access points that don't require a password - in older WPA such connections are not protected at all)
- Extra security for users with weak passwords
- Easier to connect IoT devices to WiFi networks

Fake access points

Hackers kunnen hun eigen access point opzetten

- Een access point genaamd ‘Gratis WiFi’ op een station of vliegveld
- Een ‘Evil Twin’ met hetzelfde SSID als het normale wireless access point op een locatie, maar met een sterker signal

"If there is more than one WAP with the same SSID, [...] then the client will associate with the WAP providing the strongest signal. The “Evil Twin” ruse allows any sufficiently strong 802.11 broadcaster to become a “man-in-the-middle” between the unwitting client and every other system that it communicates with. [...] Once a client has connected to the “Evil Twin,” the attacker can intercept traffic, replace images or words on the fly, conduct SSL-stripping attacks, harvest credentials, and more.“ [Davidoff & Ham, 2012]

Ook deze tools zijn kant en klaar te koop…

(Prijs inclusief een hip tasje met ‘morale patch’)

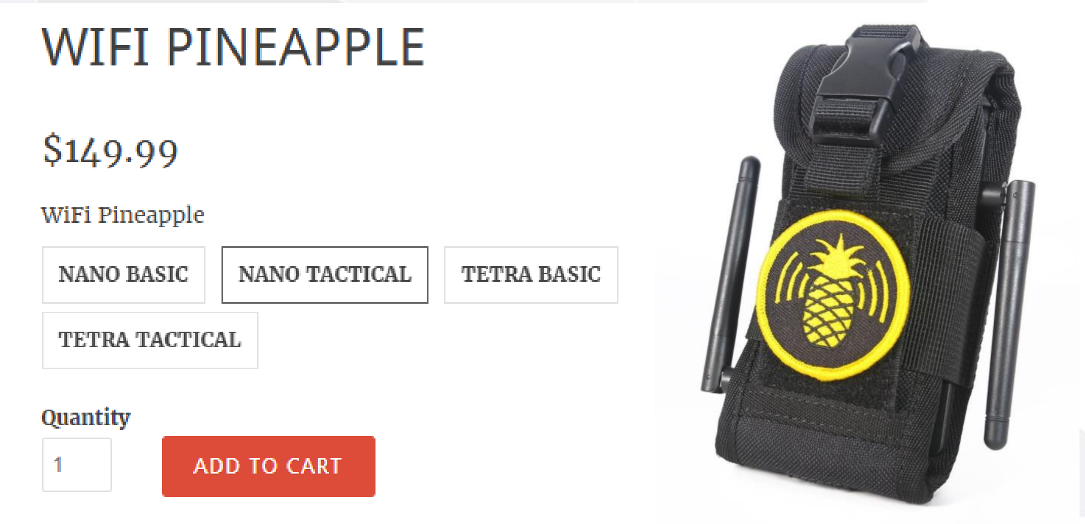

Figure 27

WiFi encryption vs HTTPS

Do we need both?

HTTPS secures the HTTP traffic between client and webserver

- but only for this one single website!
- other websites may not be secured
- and other protocols won't be secured at all…

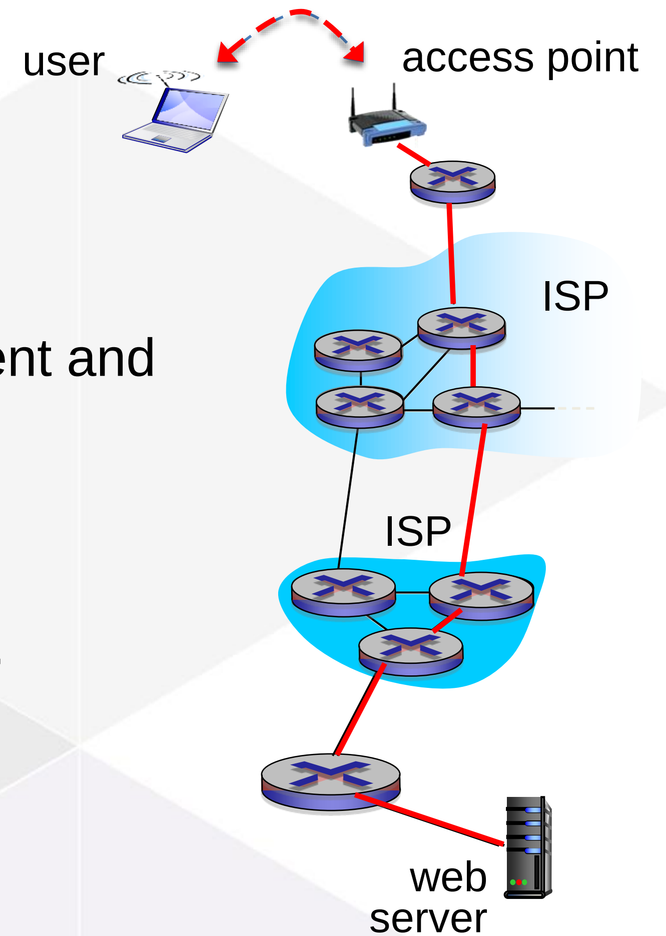

Figure 28

WiFi encryption vs HTTPS

Do we need both?

WiFi encryption just secures the connection between user and access point

- it does this for all traffic!
- but hackers could still strike elsewhere in the network

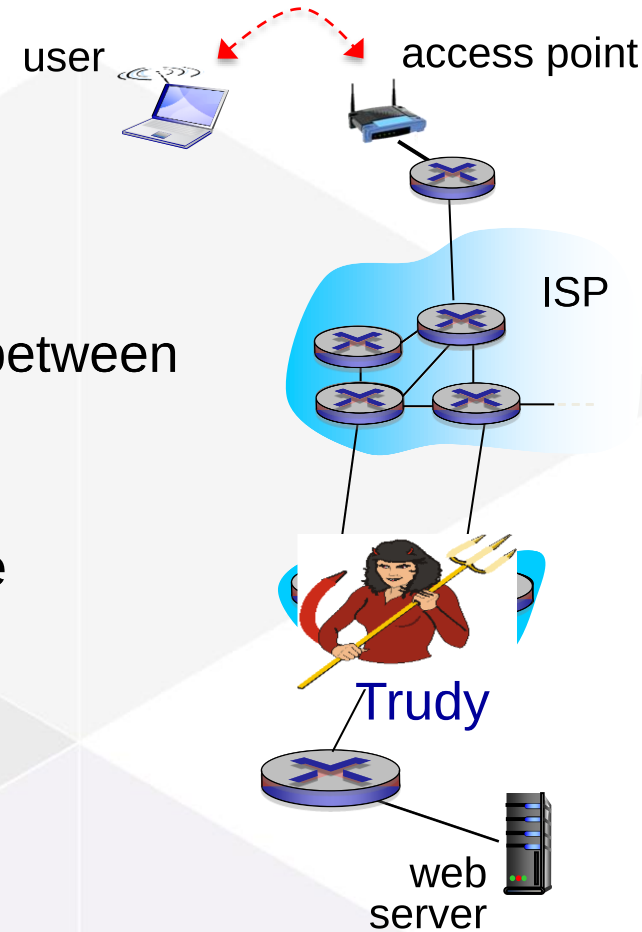

Figure 28

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Sources

Sherri Davidoff en Jonathan Ham, 2012, Network Forensics - tracking hackers through cyberspace, Prentice Hall

Figure 1,2,3,4,5,6,8,10,11,12,13,14,15,16,17,18,19,20,21,22,23, 28: Kurose and K.W. Ross, Computer Networking, A Top-Down Approach, Pearson Education, 2016

Figure 24: Home Depot, Commercial Electric 15 ft Cat5e Ethernet Cable Blue, Online Image, retrieved on 31-10-2019 from https://www.homedepot.com/p/Commercial-Electric-15-ft-Cat5e-Ethernet-Cable-Blue-575681-15/202669350

Figure 25: Riverbed Technology, AirPcap Tx USB 802.11b/g Adapter, retrieved on 27-9-2017 from https://store.riverbed.com/

Figure 26: σουμαχερ Σ, 24 oktober 2012, CommView for WiFi- How to crack WiFi WEP password, retrieved on 27-9-2017 from https://www.youtube.com/watch?v=k1kxMbddvsE

Figure 27: Hak5, WiFi pineapple, retrieved on 27-9-2017 from https://hakshop.com/products/wifi-pineapple?variant=11303796101

Figure 28: Morgan, Announcing our Worst Passwords of 2016, retrieved on 31-10-2019 from https://www.teamsid.com/worst-passwords-2016/?nabe=4561770576609280:1,5716650381017088:0,5767892520140800:2